Kernel Methods for the Incorporation of Prior-Knowledge into Support Vector Machines
–THESIS SYNOPSIS–

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May 2012

In this thesis, we present the Knowledge-Enhanced RBF (KE-RBF) framework, a family of kernel methods for the incorporation of prior-knowledge into SVMs. The KE-RBF framework consists in three different types of kernels (ξRBF, pRBF and gRBF) based on adaptations of the standard RBF kernel.

KE-RBF kernels enable the incorporation of a wide range of prior-knowledge specific to the task including global properties such as monotonicity, pseudo-periodicity or characteristic correlation patterns, and semi-global properties represented by unlabelled and labelled regions of the feature space.

The methods are numerically evaluated on several real-world applications using publicly available data. The empirical study shows that with adequate prior-knowledge, the methods are able to significantly improve the results obtained with standard kernels. In particular, they enable learning with very small or strongly biased training sets significantly broadening the field of application of SVMs.

The thesis also contains an application of the KE-RBF framework to a large-scale project for the automatic diagnosis and prognosis of breast cancer. The project involves academic, industrial and medical partners, with deployment and validation in real clinical environment.

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1 Motivation

Support Vector Methods (SVM) with their numerous variants for classification and regression tasks are state-of-the-art machine learning algorithms. Some of their key features are the absence of local optima, the possibility to control over-fitting and the use of kernels. In combination with the nonlinear Radial Basis Function (RBF) kernel, they provide a powerful and versatile learning tool often used as a default choice in many real-world applications.

SVMs only require labelled sample points as input. Therefore, as long as training data is available in sufficient quantity and quality, the SVM+RBF combination can be applied as a general-purpose learning black-box on the data and often produce good results. On complex problems, such methods can however lead to steep requirements in training data.

Unfortunately, countless reasons (cost issues, time constraints, ethical reasons, etc...) make training data for real-world problems hard to obtain. Meanwhile, real-world problems are seldom black-boxes as some general or specific knowledge about the task is often available. Thus, it seems natural to rely upon such additional *prior-knowledge* when training data is insufficient.

In this thesis, we propose the Knowledge-Enhanced RBF (KE-RBF) kernel framework, a family of kernel methods for the incorporation of prior-knowledge into SVMs. Based upon adaptations of the standard RBF kernel according to the prior-knowledge they aim at incorporating properties highly characteristic of particular problems while preserving the versatility making the popularity of the RBF kernels.

The KE-RBF framework consists of three distinct types of kernels: ξRBF kernels, pRBF kernels and gRBF kernels. Together, they enable the incorporation of a wide range of prior-knowledge specific to the task including global properties such as monotonicity, pseudo-periodicity or characteristic correlation patterns, and semi-global properties represented by unlabelled and labelled regions of the feature space.

A systematic empirical validation on several real-world applications shows that the methods are highly effective and easy to use in practice. In particular, they enable learning with very small or strongly biased training sets significantly broadening the field of application of SVMs.

This synopsis is structured as follows. In section 2, we give a brief introduction to SVMs from a statistical perspective followed by a presentation of the problem of prior-knowledge incorporation into SVMs in section 3 with an overview of the existing related works. The individual kernels composing the KE-RBF framework are presented in section 4 illustrated with relevant real-world applications examples used for empirical evaluation in the thesis. Finally, some contributions of SVMs and the KE-RBF framework to large-scale applications are presented in section 5.

The purpose of this synopsis is to provide a condensed overview of methods illustrated with some applications proposed in the thesis. Full technical details on the methods, their detailed numerical evaluation and some additional applications are available in the

thesis report.

2 Support vector methods

SVMs are a family of supervised learning methods used for pattern recognition. They are often presented from a geometrical standpoint as the construction of a hyperplane in a real Hilbert space. The hyperplane is used to separate classes or as a regression model. From a statistical standpoint, SVMs are the implementation of a statistical learning strategy first proposed by Vapnik and Chervonenkis [29] in 1974 and known as the Structural Risk Minimization (SRM) principle.

Let $(\mathcal{X} \times \mathcal{Y}, \mathcal{A}, \mathscr{P})$ be a probability space where $\mathcal{X}$ is known as the *input (or feature) space*, $\mathcal{Y} \subset \mathbb{R}$ is the *output (or label) space* and $\mathscr{P}$ is referred to as the *problem* distribution. Given a set of *labelling models* $\mathcal{H}_B \subset \mathcal{Y}^{\mathcal{X}}$ and a loss function $\Lambda : \mathcal{X} \times \mathcal{Y} \times \mathcal{H}_B \to \mathbb{R}$, the goal of statistical learning is to find the model $f \in \mathcal{H}_B$ that minimizes the *theoretical risk*, *i.e.* the average value of the loss function over $\mathscr{P}$:

$$R(f) = \mathbb{E}_{(X,Y)\sim\mathscr{P}}(\Lambda(X, Y, f)) \tag{1}$$

In practice, $\mathscr{P}$ which describes the distribution of observations is not known. Therefore, it is not possible to minimize the theoretical risk directly. Instead, $R(f)$ is approximated by the *empirical risk* $R^*(f)$ over a finite *training set* $\mathcal{S}_N = (x_i, y_i)_{i=1}^N$ of observations *i.i.d.* according to $\mathscr{P}$:

$$R^*(f) = \frac{1}{N} \sum_{i=1}^{N} \Lambda(x_i, y_i, f) \tag{2}$$

However, the labelling model f minimizing $R^*(f)$ can be far from a minimum of $R(f)$, a problem described as *over-fitting* $\mathcal{S}_N$.

The SRM principle is based upon a bounding of the theoretical risk under some hypothesis on Λ, $\mathcal{Y}$ and $\mathcal{H}_B$. In particular, $\mathcal{H}_B$ must be a ball of radius B in a Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}$. Then with "high-probability":

$$R(f) \leq R^*(f) + B\Delta \tag{3}$$

where $\Delta \geq 0$ is a constant. Therefore, rather than minimizing the empirical risk $R^*(f)$, one should strike the right balance between a minimization of $R^*(f)$ and a minimization of $B\Delta$ (hence of B) referred to as the *capacity term*.

SVMs are a direct implementation of the SRM principle. With parameter $\lambda \geq 0$ balancing the previously mentioned trade-off, SVMs solve the following problem sometimes described as the minimization of the *regularized* empirical risk:

$$\hat{f} = \underset{f\in\mathcal{H}}{\operatorname{argmin}}\ R^*(f) + \lambda\|f\|_{\mathcal{H}}^2 \tag{4}$$

The trade-off parameter λ is usually adjusted with tuning methods.

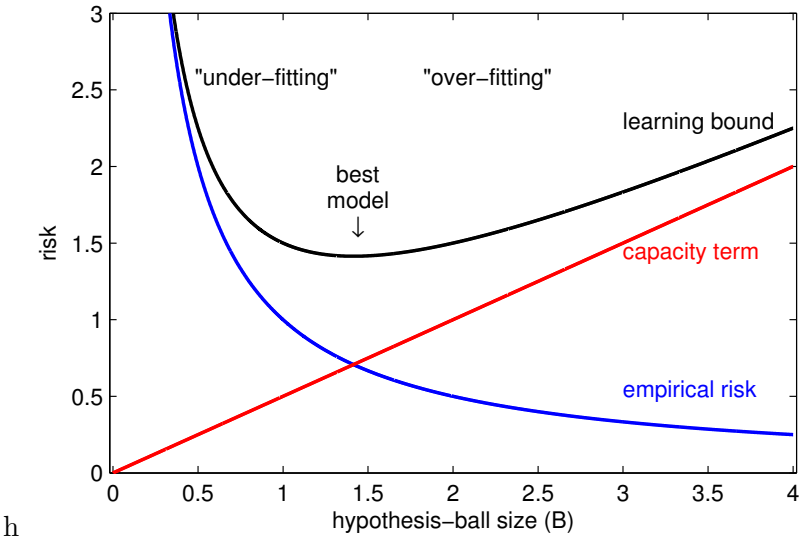


Figure 1: The theoretical risk $R(f)$ is bounded by the sum of the monotonically decreasing empirical risk $R^*(f)$ and the monotonically increasing capacity term $B\Delta$.

The different types of SVMs are usually defined by the loss function used and the way to parametrize the trade-off. Two loss functions commonly used with SVMs are the hinge loss $\Lambda_{\text{hinge}}(x, y, f) = \max(0, 1 - yf(x))$ for binary classification problems and the ϵ-insensitive loss $\Lambda_\epsilon(x, y, f) = \max(0, |y - f(x)| - \epsilon)$ for scalar regression problems.

A central result in kernel theory known as the *representer theorem* guarantees that the solution can be expressed as:

$$\hat{f}(x) = \sum_{i=1}^{N} \alpha_i K(x_i, x) \tag{5}$$

with $(\alpha_i)_{i=1}^N \in \mathbb{R}^N$ and K the Positive-Definite (PD) reproducing kernel of $\mathcal{H}$. With adequate loss functions such as the hinge loss or the ϵ-insensitive loss, (4) becomes a (sparse) convex optimization problem for which many efficient solving strategies have been developed.

By substituting (5) into (4), the problem can be entirely expressed in terms of kernel products over the training data. Therefore, SVMs only require kernel Gram matrices (and labels) as input and are consequently referred to as *kernel methods*. The choice of PD kernel K which implicitly defines the RKHS $\mathcal{H}$ from which the models are drawn considerably influences the learning results. A particular nonlinear kernel over $\mathbb{R}^n$ known as the RBF kernel defined as $K_{\text{rbf}}(x_1, x_2) = \exp(-\gamma\|x_1 - x_2\|_2^2)$ (with $\gamma > 0$ the bandwidth parameter) is widely used due to its versatility and good average results.

Remark 2.1. *Aronszajn's theorem* states that a PD kernel is the inner product in the RKHS after applying a mapping Φ to the data from $\mathcal{X}$ to $\mathcal{H}$, *i.e.* $K(x_1, x_2) = \langle \Phi(x_1), \Phi(x_2) \rangle_{\mathcal{H}}$.

Therefore, the solution (5) becomes:

$$\hat{f}(x) = \sum_{i=1}^{N} \alpha_i \langle \Phi(x_i), \Phi(x) \rangle_{\mathcal{H}} = \langle \sum_{i=1}^{N} \alpha_i \Phi(x_i), \Phi(x) \rangle_{\mathcal{H}} \tag{6}$$

which corresponds to the equation of a hyperplane in $\mathcal{H}$. Note that all hyperplanes defined in this fashion pass through the origin of $\mathcal{H}$. An offset variable $b \in \mathbb{R}$ is often added to the solution to allow for affine hyperplanes.

3 Prior-knowledge incorporation into SVMs

The incorporation of prior-knowledge into SVMs has been the focus of a number of previous works. We propose a classification of the types of prior-knowledge in 3 categories:

- domain-specific: properties such as invariances to transformations which apply to the domain (*e.g.* rotational invariance in computer vision) rather than the problem itself (*e.g.* face recognition);
- data-specific: additional information on the training data (*e.g.* class imbalance) or the unlabelled data;
- problem-specific: properties describing aspects highly characteristic of the task (*e.g.* interpretation of certain features).

Works on domain-specific knowledge are the most numerous. The majority deals with transformation invariances. The earliest example is the virtual samples method [25] consisting in the generation of artificial samples from the original ones. More recent methods relate to a substitution of the data points with their equivalence classes. This includes: the jittering kernels [12], the tangent distance kernels [15], the tangent vectors kernels [24], the Haar integration kernels [16], the π-SVM [27], the semi-definite programming machines [14] and the invariant hyperplanes [26]. Some other works incorporate notions of distances for specific types of objects (other than tuples from $\mathbb{R}^n$) such as sets of vectors [17] or local alignment kernels for sequences [31].

Additional knowledge on the data such as class imbalances can be incorporated by attributing class specific misclassification cost parameters, a method first proposed in [30], or by selecting a kernel based on the scatter of the classes in the RKHS [32]. Other methods such as the transductive SVM [28] take into account the distribution of the unlabelled data during training.

Previous works on task-specific knowledge relates to the Knowledge-Based SVM (KB-SVM) framework [22, 23] which allows for the incorporation of labelled sets obtained from expert knowledge into the problem in addition to labelled data points. Extensions of this work was later proposed including several simplified frameworks [20, 21] and an online version [18]. [13] proposed an alternative approach based on generation of artificial data samples.

The above previous works can also be classified into 3 categories according to the incorporation method. They can be:

- sample-based: additional training samples are generated from the prior-knowledge;
- kernel-based: a specific kernel is designed or selected according to the prior-knowledge;
- optimization-based: the formulation of the optimization problem is modified to incorporate the prior-knowledge, usually by adding some new constraints.

Some works may relate to several incorporation methods. A review on the subject structured according to the incorporation method is available from [19].

Table 1: Matrix of the previous work with selected references. Columns correspond to types of prior-knowledge and rows to incorporation methods. Our KE-RBF framework matches the colored cell.

	Domain-specific	Data-specific	Task-specific
Sample-based	• Virtual samples [25] • π-SVM [27]		• Knowledge initialization [13]
Kernel-based	• Jittering kernels [12] • Tangent distance kernels [15] • Tangent vector kernels [24] • Haar integration kernels [16] • Kernels for finite sets [17] • Local alignment kernel [31]	• Weighted samples [30] • Knowledge-driven kernel selection [32]	
Optimization-based	• π-SVM [27] • Semi-definite programming machines [14] • Invariant hyperplanes [26]	• Weighted samples [30] • Transductive SVM [28]	• KBSVM [22, 23] • Extensional KBSVM [21] • Simpler KBSVM [20] • Online KBSVM [18]

Table 1 gives a matrix overview of the previous work according to type of knowledge and the incorporation method.

The largest part of the previous works relate to domain-specific knowledge (mostly invariances) which can only loosely capture the properties specific to a given problem. The work on data-specific knowledge is not designed with this purpose in mind either. Comparatively little work has been done on prior-knowledge specific to the problem itself, and mostly consists in the KBSVM framework and derived methods which enable the incorporation of prior-knowledge in the form of labelled sets.

Moreover, the KBSVM-related methods are optimization-based methods: they incorporate the prior-knowledge in the form of additional constraints into the problem. In addition to being fairly complex in design, these methods do not modify the search space $\mathcal{H}$ of the solution given by (5) which directly depends on the training data and the kernel. With very small or strongly biased datasets, $\mathcal{H}$ may be either too small or too biased in order to contain an adequate solution.

We believe that a kernel-based approach is the most appropriate for our purpose. On one hand, the search space of the solution itself is modified to fit the prior-knowledge. On the other hand, the "kernel trick" is arguably the most natural approach to modifying SVMs as it does not denature the SRM principle justifying the good statistical performances of SVMs.

The KE-RBF framework presented in the following section aims at bridging the gap identified in the previous works by proposing a family of practical kernel-based methods for the incorporation of a wide variety of commonly available problem-specific knowledge into SVMs.

4 Knowledge-enhanced RBF kernels

The KE-RBF framework is a set of original methods for the incorporation of prior-knowledge into SVMs. It comprises 3 new kernels (the ξRBF, pRBF and gRBF kernels) based on transformations of the RBF kernel widely used in machine learning. The main motivation is to give systematic methods for the incorporation of problem-specific properties while retaining the versatility making the popularity of the RBF kernel.

The KE-RBF kernels allow for the incorporation of a wide array of commonly available problem-specific prior-knowledge including global properties such as monotonicity, pseudo-periodicity or characteristic correlation patterns and semi-global properties represented by unlabelled and labelled regions of the feature space. Table 2 provides a quick matrix overview of which type of kernel can be used to incorporate which type of properties.

The following sections introduce each of the kernels through examples of application on real-word problems. KE-RBF kernels are highly usable in practice and pave the way for several interesting new possibilities with SVMs such as learning with very small or strongly biased datasets.

		ξRBF	pRBF	gRBF
semi-global	unlabeled regions	×		
	labeled regions			×
global	monotonicity		×	
	pseudo-periodicity	×		
	frequency decomposition	×		
	explicit correlation		×	

Table 2: Matrix representation of the different types of KE-RBF kernels (top) with the different types of prior-knowledge (left). Crosses indicate kernels that can be used with a specific type of prior-knowledge.

4.1 ξRBF

The ξRBF kernels can be defined as the functional product of the RBF kernel with a *knowledge function* ξ:

$$K_a(x_1, x_2) = (\lambda + \mu\xi(x_1, x_2))K_{\text{rbf}}(x_1, x_2) \tag{7}$$

where $\xi : \mathcal{X}^2 \to \mathbb{R}$ is a symmetric function containing the prior-knowledge. The parameter $\mu = 1 - \lambda \in [0, 1]$ controls the amount of prior-knowledge incorporated.

The idea is to alter the kernel distance in order to influence the separability of points according to the prior-knowledge. On one hand, if the prior-knowledge suggests that two objects share similarities, then the objects should be moved closer and the kernel distance decreased. On the other hand, if it implies that those two objects are unrelated or dissimilar, they should be moved further apart and the kernel distance increased.

The knowledge function ξ is designed according to the type of prior-knowledge incorporated. In this thesis, we design knowledge functions for 2 different types of prior-knowledge:

- similarity across a region of $\mathcal{X}$ defined as crisp and fuzzy sets, without hypothesis on the label;
- information on the frequency decomposition of the labeling model *w.r.t.* one or several features.

4.1.1 Unlabelled regions

In some applications, regions $\mathcal{A} \subset \mathcal{X}$ of the feature space may be identified in advance as "sharing similarities" without additional hypothesis on the labels.

The following describes a concrete medical example used for the validation of our framework and for which detailed results are available in [9]. Fine Needle Aspiration (FNA) biopsies are a common practice for the diagnosis of breast cancer. The micrographs present a population of cell nuclei on a clear and uniform background which is observed by a pathologists who must judge if the case is benign or malignant. The diagnosis of breast cancer from cytological images is based upon the study of Cytonuclear

Atypiae (CNA), *i.e.* any feature uncharacteristic of normal nuclei. Using his knowledge in oncology, the pathologist can define regions $\mathcal{A} \subset \mathcal{X}$ corresponding to sets of typical nuclei.

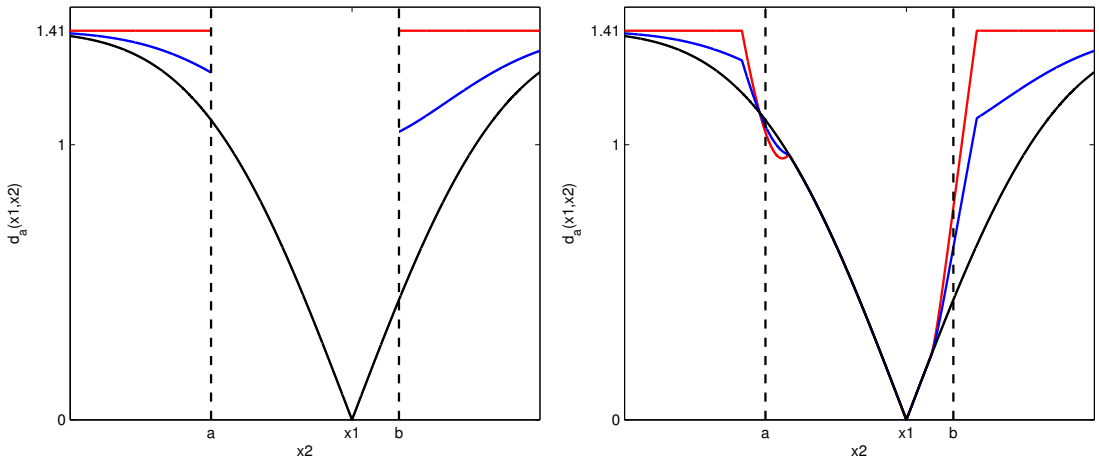
The prior-knowledge corresponding to such unlabelled sets can be incorporated using the following knowledge function:

$$\xi(x_1, x_2) = \frac{\chi(x_1)\chi(x_2) + 1}{2} \tag{8}$$

where $\chi : \mathcal{X} \to \{-1, 1\}$ is an indicator function for the set $\mathcal{A}$ such that:

$$\chi(x) = \begin{cases} 1 & \text{if } x \in \mathcal{A} \\ -1 & \text{if } x \notin \mathcal{A} \end{cases} \tag{9}$$

The resulting ξRBF kernel K_a is PD and the kernel distance d_a in the corresponding RKHS $\mathcal{H}_a$ between two points $x_1 \in \mathcal{X}$ and $x_2 \in \mathcal{X}$ can be analytically computed. Figure 2a shows plots of the kernel distance $d_a(x_1, x_2)$ in a one-dimensional example ($\mathcal{X} = \mathbb{R}$) with $\mathcal{A} = [a, b]$ and different values of the parameter $\mu \in [0, 1]$. We can observe that when the two points are in the same set, the kernel distance between them is the standard RBF kernel distance. However, when they are in different sets, the kernel distance increases by an amount controllable via the parameter μ.



(a) Crisp set (b) Fuzzy set

Figure 2: ξRBF kernel distance $d_a(x_1, x_2)$ for $\mathcal{X} = \mathbb{R}$, an unlabelled set $\mathcal{A} = [a, b]$ and different values of the parameter μ. Black plots correspond to $\mu = 0$ *i.e.* the standard RBF kernel, blue plots to $\mu = 0.5$ and red plots to $\mu = 1$.

The consequence for our histopathological application is that the separability between populations of nuclei belonging to different sets is increased whereas the separability between populations belonging to the same set will be relatively decreased, incorporating the notion of similarity.

In practice, the boundaries of the regions may not be precisely known. Instead, the prior-knowledge may correspond to a blur idea of them. In such a case a continuous indicator function $\chi : \mathcal{X} \to [-1, 1]$ defining a fuzzy set may be used. Figure 2b is a fuzzified version of the illustration in figure 2a with crisp sets. We can see that the previously discontinuous transitions are now smooth.

4.1.2 Frequency decomposition

ξRBF kernels can also be used to incorporate prior-knowledge about the expected frequency decomposition of the problem.

For instance, in a meteorological application to the prediction of atmospheric temperatures, we expect the data to follow 2 cycles:

- the cycle of seasons with a pseudo-period $P_1 = \frac{1}{f_1} = 356.25$ days.
- the diurnal cycle with a pseudo-period $P_2 = \frac{1}{f_2} = 1$ day.

In such a case where N_0 dominant frequencies can be identified, we can use the following knowledge function:

$$K_a(x_1, x_2) = \prod_{i=1}^{N_0} \xi_i(x_1, x_2) \tag{10}$$

with each ξ_i for each of the frequencies defined as:

$$\xi_i(x_1, x_2) = \frac{\cos(2\pi f_i(x_{1,j} - x_{2,j})) + 1}{2} \tag{11}$$

where $x_{1,j}$ (resp. $x_{2,j}$) is the j-th component of x_1 (resp. x_2) on which the prior-knowledge about the frequency decomposition applies.

Figure 3 shows a plot of d_a for the one-dimensional case $\mathcal{X} = \mathbb{R}$ with a single pseudo-period. We can observe a pseudo-periodic increase in the kernel distance compared to the standard RBF distance ($\mu = 0$) in addition to the exponential increase proper to the RBF kernel. Therefore, objects which are separated by a whole number of pseudo-periods are more strongly related than objects separated by a non-whole number of pseudo-periods.

In terms of the meteorological application, this could imply that a summer day and a winter day of the same year are further apart in kernel space than two summer days separated by several years.

Systematic empirical validations on the medical application presented in section 4.1.1 and the meteorological application presented section 4.1.2 show that ξRBF kernels can significantly improve results compared to the standard, specially for small training sets. Detailed experimental results for the two applications are available in [9].

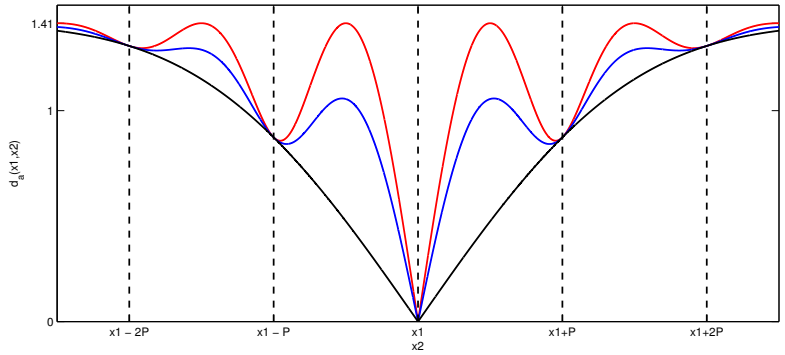


Figure 3: ξRBF kernel distance $d_a(x_1, x_2)$ for $\mathcal{X} = \mathbb{R}$ and a pseudo-periods P. The interval between dashed lines is equal to P. black plots correspond to $\mu = 0$ *i.e.* the standard RBF kernel, blue plots to $\mu = 0.5$ and red plots to $\mu = 1$.

4.2 pRBF

pRBF kernels are used to incorporate knowledge about specific correlation patterns between features and labels. The correlation can be simple such as monotonicity (*e.g.* "the financial solvency of a person increases with her income") or more specific (*e.g.* "the stopping distance of a car is quadratically correlated to its speed").

In brief, a pRBF kernel is a tensor product between the RBF kernel and another kernel K:

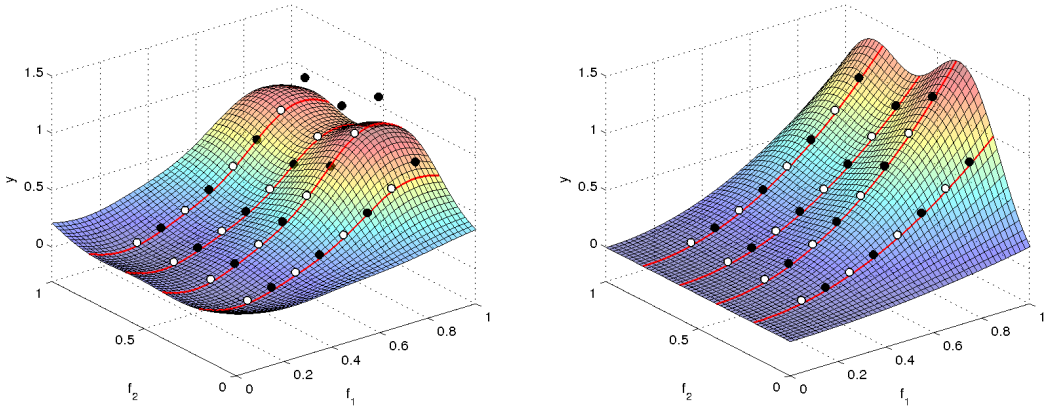
$$\begin{array}{rccl} K_a: & \mathbb{R}^n \times \mathbb{R}^n & \to & \mathbb{R} \\ & (x_1, x_2) & \mapsto & K_{\text{rbf}}(x_{1,1}, x_{2,1}) \times K(x_{1,2}, x_{2,2}) \end{array} \tag{12}$$

with $x_1 = (x_{1,1}, x_{1,2}) \in \mathbb{R}^{n-m} \times \mathbb{R}^m$, $x_2 = (x_{2,1}, x_{2,2}) \in \mathbb{R}^{n-m} \times \mathbb{R}^m$ and $m \in [\![0, n]\!]$. The non-RBF portion of the kernel is used on features for which specific prior-knowledge is available and the RBF portion of the remaining features without any specific information.

Under certain hypothesis on the non-RBF portion K, we can prove that the relevant properties of K are preserved in the solution of the SVM. In particular, certain monomial and polynomial correlations (with certain constraints on the degree) are maintained.

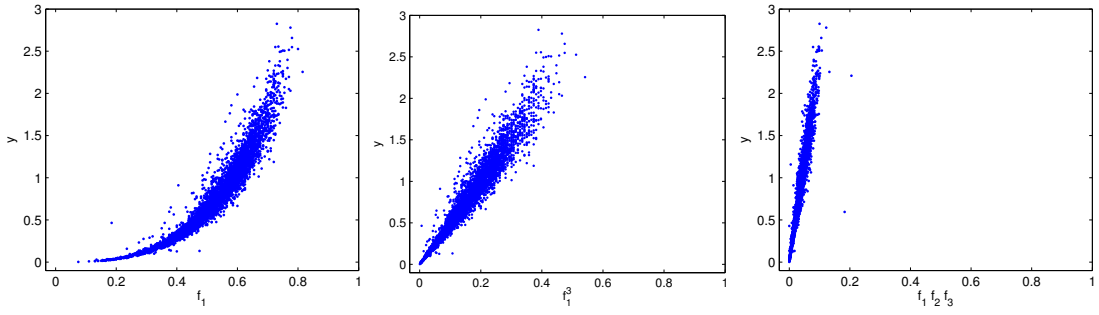
Figure 4 is a 2-dimensional illustration on a regression problem using the ϵ-SVR. In this example, the label has a quadratic correlation *w.r.t.* the fist feature f_1. When the standard RBF kernel is used (Figure 4a), the resulting decision model fits the training data (white dots) but not the test data (black dots). When a pRBF kernel is used with the correct quadratic assumption on f_1, all the test data is correctly labelled including the points out of the range of the training data. Extrapolation capabilities out of the range of the training data is usually not expected from SVMs.

Such correlation patterns occur in many real-world applications including the following one consisting in the prediction of the weight of abalones from their anatomical features including dimensions (length, diameter, height) and other features. Figure 5 representing the weight y of the 4177 abalones plotted against a few monomial combinations of the features shows that the correlation between the dimensions and the weight is monotonic and in fact cubic, as we can expect from simple geometrical intuition.



(a) RBF kernel (b) pRBF kernel

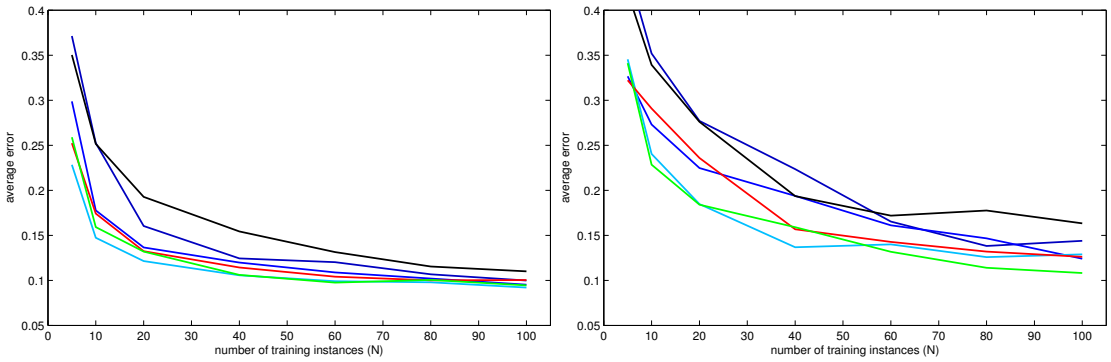
Figure 4: Regression using the ϵ-SVR+pRBF combination. The training data points are indicated with white dots and the test data points with black dots. The red curves drawn on the decision surface are level curves *w.r.t.* f_2.



(a) y against f_1. (b) y against f_1^3. (c) y against $f_1 f_2 f_3$.

Figure 5: Weight of the abalones (y) against combinations of length (f_1), diameter (f_2) and height (f_3).

Figure 6a shows a comparison of the numerical results obtained with different pRBF kernels and the standard RBF kernel. The training sets have been obtained by randomly selecting training instances in an unbiased fashion. pRBF kernels systematically improve the results obtained with the standard RBF kernel specially for small training sets. The improvements are more significant when more accurate prior-assumptions are made (cubic correlation). Figure 6b shows empirical results when biased training sets are used (only infant abalones were used for training). A noticeable difference with the unbiased case is that the improvements remain substantial even when the training sets become larger which confirms the extrapolation capabilities of pRBF kernels. More specific details on the empirical validation are available in [11].



(a) Unbiased training

(b) Biased training

Figure 6: Empirical results for the prediction of the weight of the abalones from anatomical features in terms of average prediction error. Black corresponds to the standard RBF kernel, dark blue (f_1) to a pRBF kernel assuming a degree 1 correlation, blue (f_1^2) and red ($f_1 f_2$) a degree 2 correlation, and light blue (f_1^3) and green ($f_1 f_2 f_3$) a degree 3 correlation.

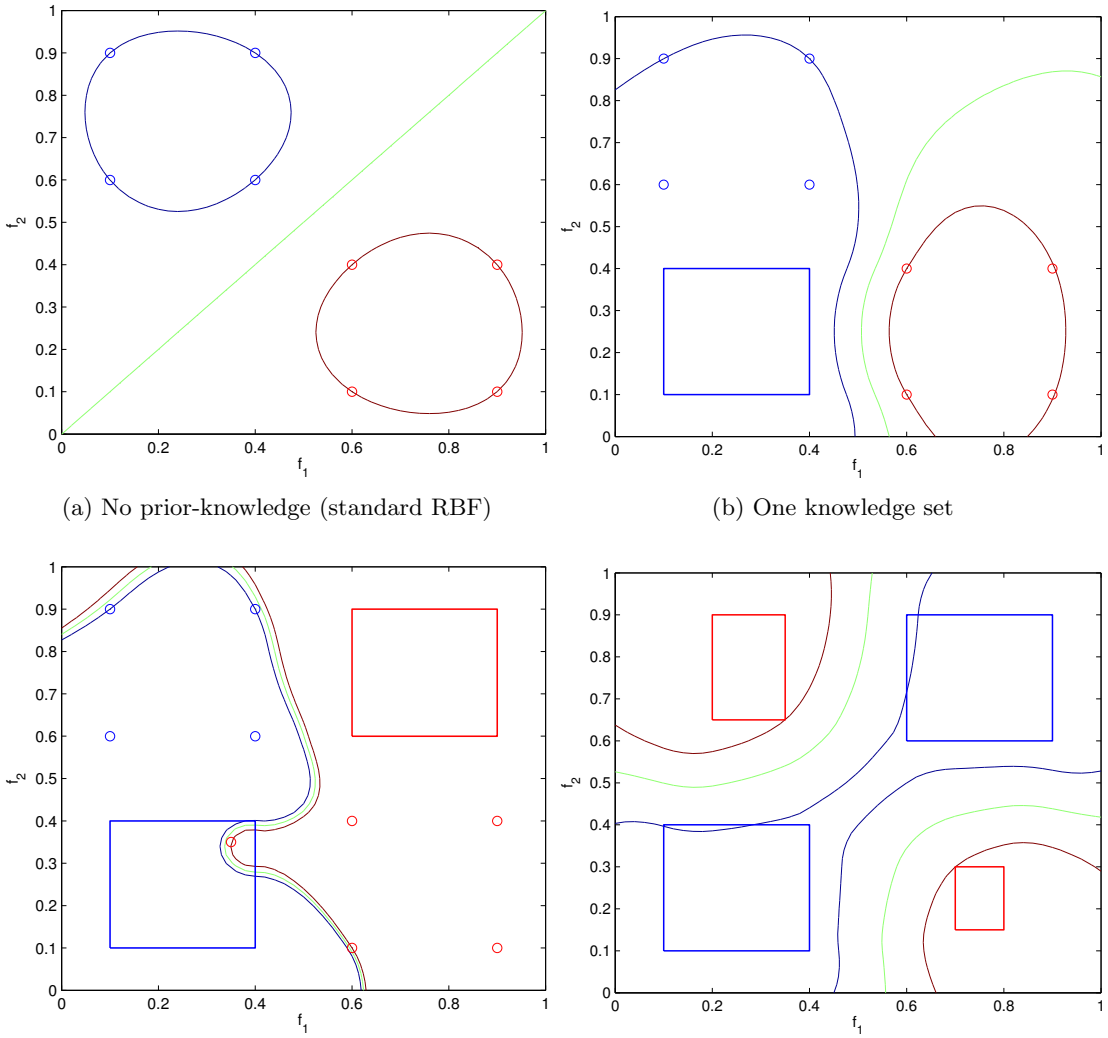
4.3 gRBF

gRBF kernels are a generalization of the standard RBF kernel from $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ to $\mathfrak{P}(\mathbb{R}^n) \times \mathfrak{P}(\mathbb{R}^n) \to \mathbb{R}$, *i.e.* from points of the feature space to sets of the feature space. Formally, the gRBF kernel is obtained by substituting the usual Euclidean distance in the expression of standard RBF kernel with a notion of set distance defined between two sets $\mathcal{A} \in \mathfrak{P}(\mathbb{R}^n)$ and $\mathcal{B} \in \mathfrak{P}(\mathbb{R}^n)$ as:

$$d(\mathcal{A}, \mathcal{B}) = \begin{cases} \inf_{a \in \mathcal{A} \wedge b \in \mathcal{B}} \|a - b\|_2 & \text{if } \mathcal{A} \neq \emptyset \text{ and } \mathcal{B} \neq \emptyset \\ \infty & \text{otherwise} \end{cases} \tag{13}$$

The gRBF kernel allows the incorporation of prior-knowledge as labelled sets into the problem. The usual labelled data points (which become labelled singletons) and other labelled sets from prior-knowledge are merged into a single training set and treated without distinction.

Figure 7 illustrates the effect of the gRBF kernel on a few two-dimensional classification examples. The examples show that the labelled sets are treated as natural extensions of the notion of labelled training sample. An noteworthy feature of the gRBF kernel is that it allows training from prior-knowledge alone without any training samples.



(a) No prior-knowledge (standard RBF)

(b) One knowledge set

(c) Two knowledge sets with a conflict between data and prior-knowledge

(d) Only prior-knowledge

Figure 7: Examples of binary classification using the C-SVM and the gRBF kernel. The red and blue circles represent the training data from the different classes. The red and blue boxes indicate the labeled regions belonging to the corresponding classes. The green line indicates the decision boundary and the red and blue lines the SVM margin.

Several challenges of a computational order are created by the use of gRBF kernels

including: the indefiniteness of the gRBF kernel (*i.e.* gRBF kernels are non-PD), the difficulty to compute the set distance for arbitrary sets, the possible conflicts between training data and prior-knowledge, and the overhead in computational complexity compared with the RBF kernel. Effective and practical solutions are proposed for each of the challenges.

An application of the gRBF kernel to meteorological predictions is also proposed to illustrate the use of labelled sets to incorporate prior-knowledge about periodical (monthly, seasonal, yearly) average values.

5 Applications

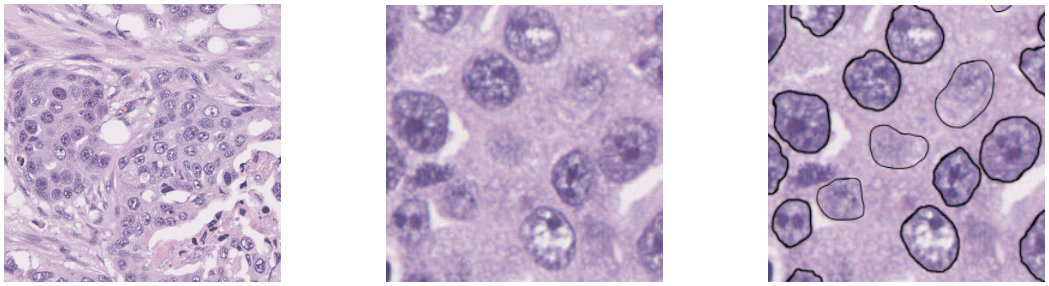
The original contributions of this thesis are applied in the context of the Cognitive Microscope (MICO) project in which they play an important role. MICO is an ongoing initiative funded by Agence Nationale de la Recherche (a French institution tasked with funding scientific research) and involving academic research laboratories[1][2], industrial partners[3][4][5] and pathologists from a university hospital[6].

MICO is a virtual microscopy platform aimed at automating a standard procedure known as Breast Cancer Grading (BCG) for the diagnosis and prognosis of breast cancer from surgical biopsy slides. Details on early work on a cognitive microscopic platform for BCG is available in [4] and [3].

Standard BCG procedures incorporate the assessment of CNAs (see section 4.1) as a central component. Therefore, algorithms able to precisely extract the cell nuclei are a strong requirement in computer-aided diagnosis applications. However, unlike other modalities such as FNA biopsy images, H&E stained surgical breast cancer slides are particularly challenging due to the heterogeneity of both the objects and background, the low object-background contrast and the frequent overlaps as illustrated in figure 8. As a consequence, existing extraction methods which are largely reliant on color intensities perform poorly on such images.

To solve the problem, we propose a method based on the creation of a new image modality consisting in a grayscale map where the value of each pixel indicates its probability of belonging to a cell nuclei. First, features based on color, texture, scale and shape information are computed for each pixel in the image. Then, a combination of SVMs and KE-RBF kernels are used to classify the pixels as belonging to a nuclei or to the background. The probability estimate for each pixels are then obtained by scaling the SVM decision function. An earlier version of the method involving Fisher's discriminant analysis instead of SVMs is available in [10].

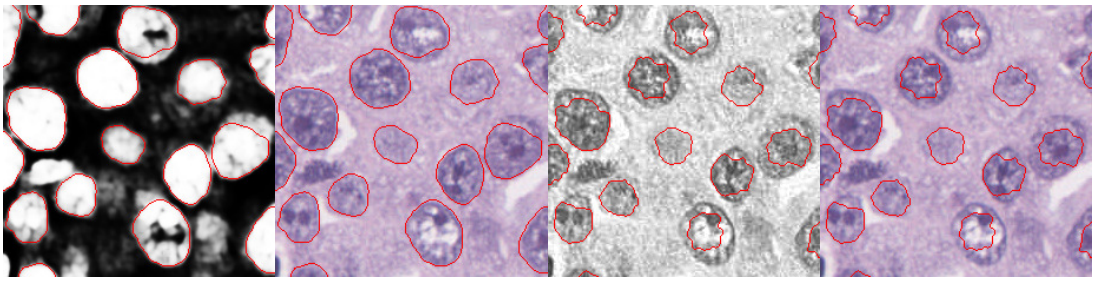
[1] Image and Pervasive Access Lab (IPAL), Université Joseph Fourier, Grenoble, France
[2] Laboratoire d'Informatique de Paris 6 (LIP6), Université Pierre et Marie Curie, Paris, France
[3] Thales Communications & Security, France
[4] AGFA-HealthCare, Belgium
[5] TRIBVN, France
[6] Groupement Hospitalier Universitaire de la Pitié-Salpêtrière (GHU-PS), Université Pierre et Marie Curie, Paris, France



(a) (b) (c)

Figure 8: (a) 1024 × 1024 pixel hyperfield, (b) magnified 250 × 250 region and (c) the same region with the nuclei delineated by a pathologist. Nuclei delineated with a thinner outline are hard to distinguish from the background. Dark and bright areas can indiscriminately occur inside and outside nuclei.

The actual extraction is performed using an AC model with a nuclei shape prior included to deal with overlapping nuclei which according to our benchmarks [2] performs the best among the state-of-the-art methods. Figure 9, shows that the new image modality significantly improves the quality of extraction.



(a) Probability map (b) Corresponding H&E stained image (c) Haematoxilin image (d) Corresponding H&E stained image

Figure 9: Comparison of extraction results : (a-b) is obtained using the probability map and (c-d) is obtained using the haematoxilin channel after the image color deconvolution.

The student also contributed to another key aspect of the MICO project by developing a method based on computational geometry for the exploration of very large images for which details are available in [6], [1] and [5].

Other activities undertaken by the student during the course of his Ph.D. studies include works on the use of SVMs for sentence-level machine translation evaluation available in [8] and [7].

6 Conclusion

The KE-RBF kernels proposed in this thesis provide tools for the incorporation of a variety of commonly available prior-knowledge into SVMs which are both practical and effective.

Their systematic evaluation on five different applications using publicly available real-world data (an synthetic data in a lesser extent) showed that KE-RBF kernels can lead to significant performance improvements when used with adequate prior-knowledge in place of the standard RBF kernel which they are able to overperform using training sets up to ten times smaller.

The performance improvements were particularly pronounced with very small or strongly biased training sets. This remarkable reduction in training data requirements enabled by the KE-RBF kernels, both quantitatively and qualitatively, opens new perspectives for SVMs significantly broadening their usual field of application.

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